

Diksha Damahe

B.Tech Computer Science & Engineering (AI & ML) | +91 97709 25370 | dikshadamahe1@gmail.com
[LinkedIn](#) | [GitHub](#) | [ORCID](#) | [Scopus ID](#)

Summary

Computer Science student specializing in Artificial Intelligence and Machine Learning, with hands-on experience in full-stack engineering, ML model pipelines, and process automation. Built and deployed responsive, secure web platforms and cross-platform mobile apps with React, Next.js, Django, and Flutter. Passionate about automating complex workflows, optimizing codebases, and creating data-driven architectures.

Skills

Languages: Python · TypeScript · JavaScript · Java · SQL

Frameworks & Libraries: React.js · Next.js · Django REST Framework · Node.js · Flask · Flutter · Tailwind CSS

Cloud & DevOps: AWS (EC2, S3, EBS) · Git · GitHub · CI/CD Workflows · Postman · Render

Databases: PostgreSQL · MongoDB · SQLite

Data & AI/ML: scikit-learn · XGBoost · SHAP · Isolation Forest · Pandas · NumPy · Feature Engineering

Experience

Hindustan Copper Limited (HCL)

Software Engineering Intern (AI/ML)

Malanjkhanda, India

May 2026 – June 2026

- Developed and deployed **RecovAI**, an AI-based predictive analytics and industrial process monitoring system for a flotation concentrator plant.
- Engineered anomaly detection pipelines and recommended optimal chemical dosages based on feed parameters, providing operators with data-driven decision support.

Projects

RecovAI (HCL Flotation Optimizer)

May 2026 – June 2026

Lead Developer · AI & Backend Pipelines

- Solo-designed and trained an **XGBoost** regression model to forecast recovery rates with an R^2 of **0.9720** and MAE of **0.1042%** based on 18+ months of real metallurgical data.
- Developed a Reagent Dose Intelligence Engine to optimize SIPX, Frother, and pH levels, integrating an **Isolation Forest** model to detect tailing process anomalies.
- Constructed a high-performance **FastAPI backend** routing predictive engines and serving an interactive Streamlit/web telemetry UI.
- Configured automated SQL database backups, Model Tracking on Render, and integrated **Groq API** endpoints for LLM-powered shift summaries.
- [GitHub](#) | [Live Demo](#)

Veil Android

April 2026 - Present

Cross-Platform Mobile Application

- Co-developed a minimalist Android streaming aggregator app utilizing **Flutter** (Dart) and **TypeScript** (resolver API).
- Integrated TMDB REST APIs for rich metadata parsing and mapped custom video layouts featuring **ExoPlayer** HLS playback.
- Integrated a self-hosted custom resolver ([cinopro-org/core](#) OMSS API) running on **Oracle Cloud (OCI)** to dynamically resolve streams.
- Shipped over **50+ CI/CD releases** using automated GitHub Actions pipelines to streamline staging and QA processes.
- [GitHub](#)

Chemical Equipment Parameter Visualizer

February 2026

Hybrid Web + Desktop Application

- Engineered a hybrid parameter visualization suite using **Django REST Framework** for the backend and **React** for the frontend.
- Built clean, responsive telemetry dashboards with **Chart.js**, rendering dynamic charts mapping chemical pressure, temperature, and flows.
- Deployed the hybrid React app to production on Vercel while packing a standalone **PyQt5** desktop client executable for offline access.
- [GitHub](#) | [Live Demo](#)

Aegis Vault

October 2025

Secure Full-Stack Password Manager

- Built a dark, glassmorphic password manager featuring client-side zero-knowledge encryption algorithms.
- Programmed a highly interactive frontend with silky transitions using **Next.js**, **TypeScript**, **Tailwind CSS**, and **Framer Motion**.
- Designed secure credentials management and authentication with **NextAuth** on a **Prisma-backed MongoDB** stack.
- [GitHub](#) | [Live Demo](#)

Publications

SPARTA: Phishing Website Detection

IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS)

Bhopal, India

- Developed the **Secure Phishing Analysis and Robust Training Algorithm (SPARTA)**, achieving a 99.15% detection rate.
- Engineered an end-to-end ML pipeline with **scikit-learn**, a **Flask REST API** backend, and a real-time URL analysis Chrome extension.

Certifications

- **Applied Machine Learning in Python** – University of Michigan (Nov 2025)
- **AWS Certification Course** – Intellipaat (Sep 2025)
- **Mastery in JavaScript** – Lernx (Aug 2025)
- **Advanced SQL for Technical Interviews** – Bosscoder Academy (Feb 2025)
- **Cloud Computing** – Elite NPTEL Online Certification, IIT Kharagpur (Jan–Apr 2025)

Education

VIT Bhopal University

B.Tech CSE (Artificial Intelligence & Machine Learning) | CGPA: 8.06

Bhopal, India

2023 – 2027